

Nikos Pagonas

Contact

Email: nikospagonas00@gmail.com — Website: nikpag.github.io

Education

Columbia University

PhD in Computer Science

Advisor: Prof. [Kostis Kaffes](#)

GPA: 3.94/4.00

Member of [DAPLab](#), focusing on improving the performance and efficiency of agentic serving. I build systems that exploit workflow structure and runtime characteristics to optimize model routing, scheduling, and resource allocation.

Sep 2024 – Present

New York, NY, US

National Technical University of Athens (NTUA)

Master of Electrical and Computer Engineering

Advisors: Prof. [Nikos Vasilakis](#), Prof. [Georgios Goumas](#)

GPA: 9.39/10.00 (14th out of 344 students)

Thesis: “*SPLASH: Scaling Out Shell Scripts on Serverless Platforms*”

Oct 2018 – Jun 2024

Athens, Greece

Experience

Google

Student Researcher

Supervisor: Prof. [Arvind Krishnamurthy](#)

Designed and implemented [Cortex](#), a prototype agentic serving platform that leverages workflow-awareness and stage isolation to improve scheduling and KV cache efficiency in agentic NL2SQL workflows.

May 2025 – Aug 2025

Sunnyvale, CA, US

Brown University

Student Researcher

Advisor: Prof. [Nikos Vasilakis](#)

Contributed to [PASH](#), an open-source [Linux Foundation](#) project. Worked on improving the performance of shell scripts by running them on serverless infrastructure, and providing fault tolerance for distributed shell script execution.

Sep 2023 – May 2024

Providence, RI, US

Arrikto Inc.

Software Engineering Intern

Supervisor: [Vangelis Koukis](#)

Explored how to serve PyTorch models more efficiently through KServe, a highly scalable machine learning deployment platform for Kubernetes.

Dec 2022 – Mar 2023

Athens, Greece

Awards	Columbia University Presidential Fellowship	2024 – Present
	A.G. Leventis Foundation Scholarship	2025 – Present
	Gerondelis Foundation Scholarship	2025
	ACM SOSP 2025 Student Travel Grant	2025
Publications	Agentic Data Environments Elaine Ang, Chenxi Huang, Georgios Liargkovas, Jerry Liu, Jinhui Liu, Nikos Pagonas, Charlie Summers, Haonan Wang, Jiakai Xu, Tianle Zhou, Yusen Zhang, Zhou Yu, Zhuo Zhang, Tianyi Peng, Kostis Kaffes, Eugene Wu <i>IEEE Data Engineering Bulletin</i> , March 2026	
	AgentOpt v0.1 Technical Report: Client-Side Optimization for LLM-Based Agents Wenyue Hua, Sripad Karne, Qian Xie, Armaan Agrawal, Nikos Pagonas, Kostis Kaffes, Tianyi Peng	
	VineLM: Trie-Based Fine-Grained Control for Agentic Workflows Nikos Pagonas, Matthew Lou, Tianyi Peng, Dan Rubenstein, Kostis Kaffes	
	Cortex: Workflow-Aware Resource Pooling and Scheduling for Agentic Serving Nikos Pagonas, Yeounoh Chung, Kostis Kaffes, Arvind Krishnamurthy <i>SOSP 2025 Workshop on Systems for Agentic AI (SAA 2025)</i>	
	λeash: Scaling Out Shell Scripts with Reliable Cloud Computing (Under Submission)	
Service	EuroSys 2026 Artifact Evaluation Committee	2026
	SOSP 2025 Artifact Evaluation Committee	2025
	OSDI 2025 Artifact Evaluation Committee	2025
	ATC 2025 Artifact Evaluation Committee	2025
	EuroSec 2024 Web and Publication Chair	2024
Teaching	Columbia University <i>Teaching Assistant</i> Course: Networks, Crowds, and the Web	Fall 2025 New York, NY, US

Professor: [Augustin Chaintreau](#)

Columbia University

Teaching Assistant

Course: [Agentic Systems Made Real](#)

Professors: [Eugene Wu](#), [Kostis Kaffes](#)

Spring 2025
New York, NY, US

Brown University

Teaching Assistant

Course: [Systems Transforming Systems](#)

Professor: [Nikos Vasilakis](#)

Fall 2024
Providence, RI, US

Guided students through a systems research project, including implementation, evaluation, and paper writing. Paper submitted to USENIX ATC 2025.

Brown University

Teaching Assistant

Course: [Distributed Systems](#)

Professor: [Nikos Vasilakis](#)

Spring 2024
Providence, RI, US

Helped revamp the course by redesigning its semester-long project and implementing its autograding infrastructure from scratch.

National Technical University of Athens

Lab Assistant

Course: Operating Systems

Professors: [N. Koziris](#), [G. Goumas](#)

Spring 2022
Athens, Greece

National Technical University of Athens

Lab Assistant

Course: Programming Techniques

Professors: [N. Papaspyrou](#), [A. Pagourtzis](#), [G. Stamou](#), [G. Goumas](#)

Spring 2020
Athens, Greece

National Technical University of Athens

Lab Assistant

Course: Introduction to Programming

Professors: [N. Papaspyrou](#), [D. Fotakis](#), [S. Zachos](#)

Fall 2019
Athens, Greece

**Research
Mentoring**

[Grace Li](#)

Columbia University, BS in CS and Math

Sep 2025 – Present

[Matthew Lou](#)

Columbia University, BS in CS and Applied Math

Sep 2025 – Present

[Joey Huang](#)

Columbia University, BS in CS

Sep 2025 – Dec 2025

Skills	Prog. Languages	Python, C, Shell, Javascript, Java, Go
	Frameworks	LangChain, LangGraph
	Platforms	AWS, Docker, Kubernetes, Apache Kafka, Apache Spark
	Tools	Git, LaTeX
Languages	English	Fluent — TOEFL iBT 116/120 (C2)
	Greek	Native